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The underwater sea bed is the new habitat where the discovery and development of Marine Pharmaceuticals are in peak. Anticancerous, Antimalarial drugs and other bioactive molecules have been isolated and tested successfully.

### 2.3.2 Phylum: Cnidaria

(G. *knode* -needle or sting cells)

Cnidarians (were previously called Coelenterata), are aquatic, sessile or free swimming, solitary or colonial forms with radial symmetry except for sea anemones (bilateral symmetry). The name Cnidaria is derived from **cnidocytes** or **cnidoblasts** with stinging cells or nematocyst on tentacles. Cnidoblasts are used for anchorage, defense, and to capture the prey. Cnidarians are the first group of animals to exhibit tissue level organisation and are diploblastic. They have a central vascular cavity or coelenteron (serves both digestion and circulatory function) with a single opening called mouth or hypostome, which serves the process of ingestion and egestion. Digestion is both extracellular and intracellular. The nervous system is primitive and is formed of diffused nerve net. Cnidarians like corals have a skeleton made up of calcium carbonate. Cnidarians exhibit two basic body forms, polyp and medusa. The polyp forms are sessile and cylindrical (e.g. *Hydra, Adamsia*), whereas the medusa are umbrella shaped and free swimming. Cnidarians which exist in both forms, also exhibit alternation of generations in their life cycle (**Metagenesis**). The polyp represents the asexual generation and medusa represents the sexual generation. Polyps produce medusa asexually and medusa forms polyps sexually. Development is indirect and includes a free swimming ciliated **planula larva**.

Examples: *Physalia* (Portugese man of war), *Adamsia* (sea anemone), Pennatula (sea pen), *Meandrina* (brain coral) (Figure 2.10).

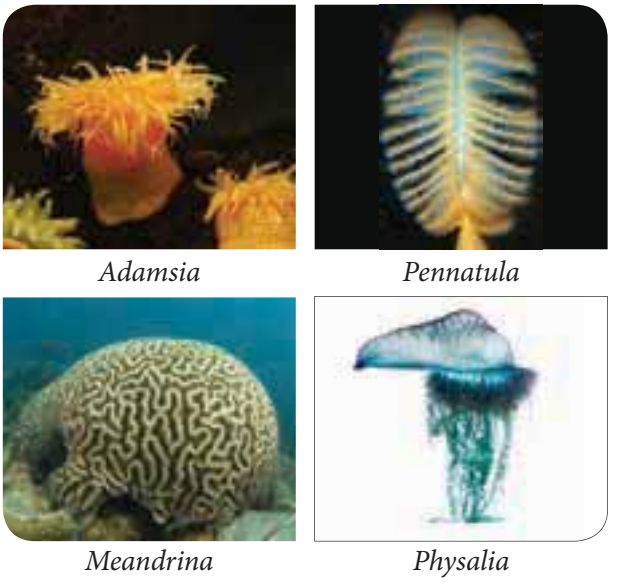


**Figure 2.10** Examples of Cnidarians

Compare the advantages and disadvantage of direct and indirect development.

### 2.3.3 Phylum: Ctenophora

(G. *Ktenos* -comb; *phoros* -bearing)

Ctenophora are exclusively marine, radially symmetrical, diploblastic animals with tissue level of organisation. Though they are diploblastic, their mesoglea is different from that of cnidaria. It contains amoebocytes and smooth muscle cells. They have eight external rows of ciliated comb plates (comb jellies) which